

Foundations of Artificial Intelligence

Intelligent Agents, Rationality, and the AI Paradigm

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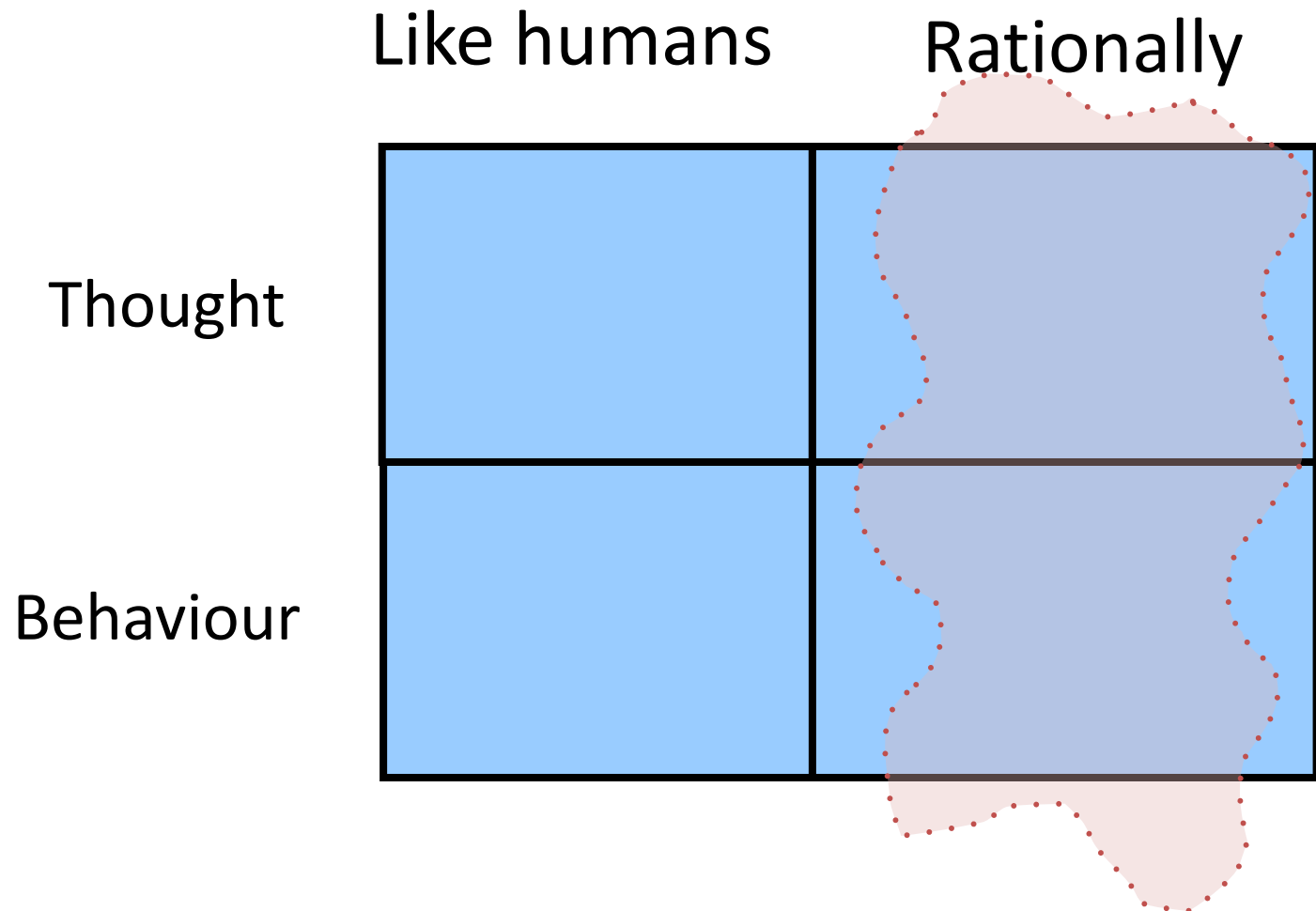
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Acknowledgement

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Possible Approaches



Thinking Humanly

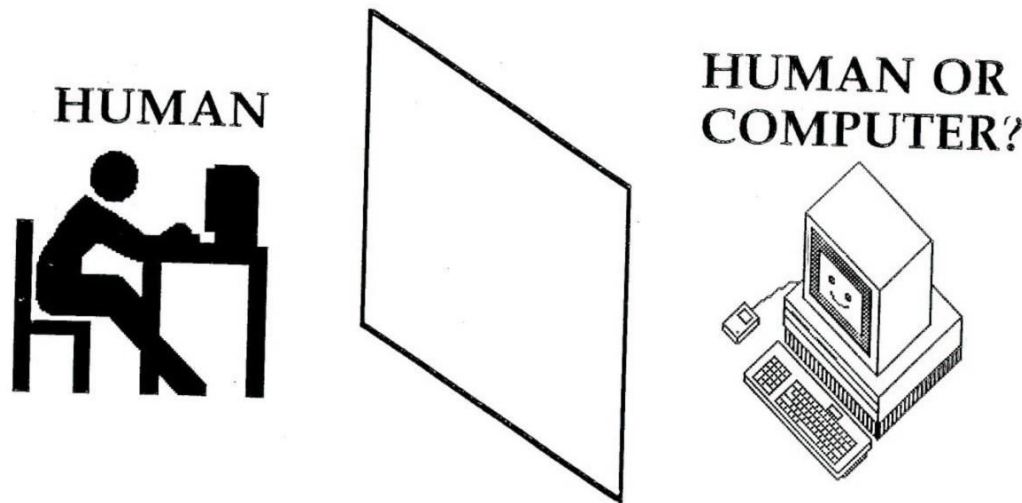
- Cognitive Science
 - Very hard to understand how humans think
- Do we want a machine that beats humans in chess or a machine that *thinks like humans* while beating humans in chess?
 - Deep Blue supposedly DOESN'T think like humans..
- Thinking like humans important in applications
 - Intelligent tutoring
 - Expressing emotions in interfaces... HCI

Thinking Rationally: laws of thought

- Aristotle: what are correct arguments/thought processes?
 - Logic
- Develop formal models of knowledge representation, reasoning, learning, memory, and problem solving, that can be rendered in algorithms.
- There is often an emphasis on systems that are provably correct, and guarantee finding an optimal solution.

Acting Humanly: Turing's Test (1950)

If the human cannot tell whether the responses from the other side of a wall are coming from a human or computer, then the computer is intelligent.



The beauty of the Turing test is its simplicity and its objectivity, because it is only a test of behavior, not of the internals of the machine. It doesn't care whether the machine is using logical methods or neural networks. The decoupling of what to solve from how to solve.

Acting Humanly

Q: Please write me a sonnet on the subject of the Forth Bridge.

A: Count me out on this one. I never could write poetry.

Q: Add 34957 to 70764.

A: (Pause about 30 seconds and then give as answer) 105621.

- What if people call a human a machine?
 - Shakespeare expert
 - Make human-like errors
- Problems
 - Not reproducible, constructive or mathematically analyzable

Imitate Humans?

- Is imitating humans the right metric?
 - Humans good at (language, vision, motor control)
 - Machines possess many advantages over humans (e.g., speed).
- Why settle for human-level performance?
 - The study of how humans think is fascinating and is well-studied within the field of cognitive science.
- In this class, however, we will primarily be concerned with the engineering goal of building intelligent systems, drawing from humans only as a source of tasks and high-level motivation.

Acting rationally

- Rational behavior: doing the right thing
- Need not always be deliberative
 - Reflexive
- Aristotle (Nicomachean Ethics)
 - Every art and every inquiry, and similarly every action and every pursuit is thought to aim at some good.

Acting ☐ Thinking?

- Weak AI Hypothesis vs. Strong AI hypothesis
 - Weak Hyp: machines could act as if they are intelligent
 - Strong Hyp: machines that act intelligent have to think intelligently too

Rational Agents

- An agent should strive to do the right thing, based on what it can perceive and the actions it can perform.
- Performance measure: An objective criterion for success of an agent's behaviour
- E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

Ideal Rational Agent

“For each possible percept sequence, does whatever action is expected to maximize its performance measure on the basis of evidence perceived so far and built-in knowledge.”

- Rationality vs omniscience?
- Acting in order to obtain valuable information

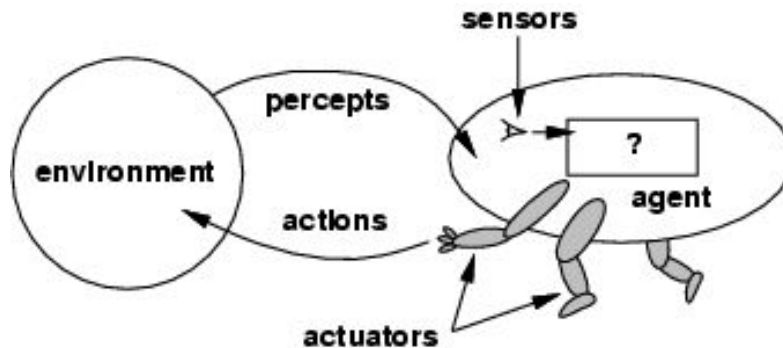
What is artificial intelligence (agent view)

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators
- Human agent:
 - Sensors
 - Actuators
- Robotic agent:
 - Sensors
 - Actuators

Autonomous Systems

- In the 1990s there was a growing concern that work in classical AI ignored crucial scientific questions:
 - How do we **integrate the components** of intelligence (*e.g.* learning & planning)?
 - How does **perception** interact with reasoning?
 - How does the demand for real-time performance in a complex, changing environment affect the architecture of intelligence?

Agents and environments



Examples of Agents

- The **agent function** maps from percept histories to actions:

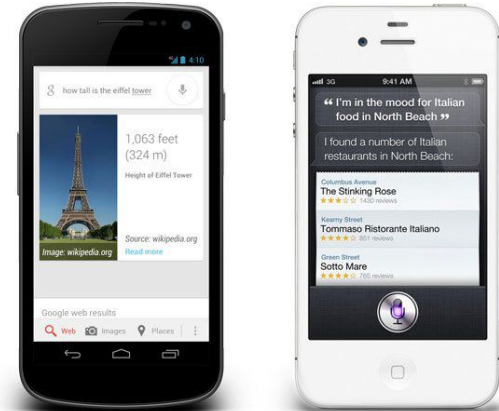
$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$

- The **agent program** runs on the physical **architecture** to produce f
- agent = architecture + program

- Robots
- Intelligent buildings
- Autonomous spacecraft
- Web agents

Examples of Agents

- Robots
- Intelligent buildings
- Autonomous spacecraft
- Self-driving cars
- Web agents
- Virtual Assistants



What is *artificial* intelligence (algorithmic view)

- A large number of problems are NP hard
- AI develops a set of tools, heuristics, ...
 - to solve such problems in practice
 - for naturally occurring instances
- Search
- Game Playing
- Planning
- ...

Example Tasks

Mundane tasks

- Perception
 - Vision
 - Speech
- Natural Language
 - Understanding
 - Generation
- Translation
 - Reasoning
 - Robot Control

Formal Tasks

- Games
 - Chess
 - Checkers
 - Othello
- Mathematics
 - Logic
 - Geometry
 - Calculus
 - Proving properties of programs

Expert Tasks

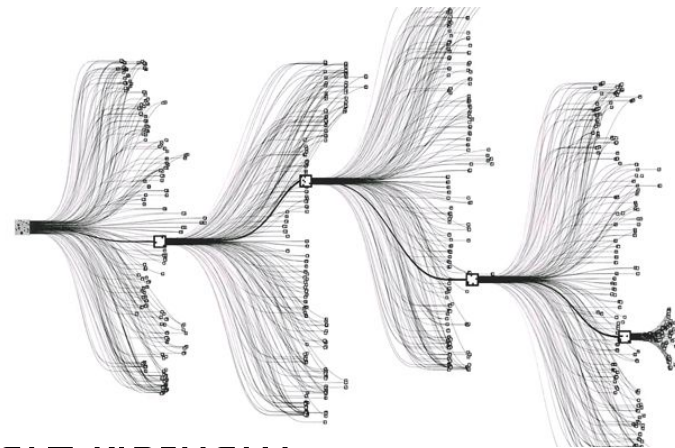
- Engineering
 - Design
 - Fault Finding
 - Manufacturing planning
- Medical
 - Diagnosis
 - Medical Image Analysis
- Financial
 - Stock market predictions

Two sources of complexity...

- Machine translation: number of possible translations?
- Input: C'est ne pas une pipe.
- Output:

**Computational complexity:
exponential explosion**

- (Vocabulary size)^(Sentence length)
- Most interesting AI problems are NP-HARD.



Information complexity: need to acquire knowledge

- Even infinite computation isn't enough...need to somehow know
 - knowledge about a foreign language, ornithology, or about what's in the dealer's deck to make optimal decisions
- **Information complexity**: uncertainty is pervasive
- But we will be always information-limited and therefore have to simply cope with uncertainty

Resources

- Computation
(time/memory)



- Information (data)



Fortunately, for AI, in the last two decades, the amount of computing power and data has skyrocketed, and this trend coincides with our ability to solve some of the challenging tasks that we discussed earlier.

- How do we tackle challenging AI tasks?

Paradigm

Break complex tasks into two stages.

1. **Modeling**: messy real-world tasks are converted into clean formal tasks called models.
2. **Algorithms**: we find efficient ways to solve these formal tasks.

Real world Task



Modelling

Formal task (Model)



Algorithms

Program



Modelling and algorithms

- Separate **what** to compute (**modelling**) from **how** to compute it (**algorithms**)
- Advantage: division of labour
- This class: providing a toolbox of different classes of models and associated algorithms

Course Topics

